MIL-A-22336B(OS)
23 March 1979
Superseding
MIL-A-22336A(OS)
21 January 1970
with
AMENDMENT 2
27 October 1972

MILITARY SPECIFICATION

# AMPLIFIER, SYNCHRO SERVO, 6 WATT 60 HERTZ

This specification is approved for use by the Naval Sea systems Command (OS, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement of a 6 watt (W), 60 Hertz (Hz] synchro servo amplifier, LD 499687, which is designed to drive a size 15, 60 cycle servo motor or a size 15, 60 Hz synchronous motor. The amplifier consists of two transistorized push-pull stages driven by the output of an incorporated transistorized pre-amplifier.

1.2 Classification. The amplifier shall be of the following type:

| | |
|---|---|
| Type I - LD 499687 | 60 Hz 6 W Synchro Servo Amplifier Assembly |

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, addoitions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (501), Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD) Form 1426] appearing at the end of this document or by letter.

FSC 1290

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-591 | Boxes; Fiberboard, Wood-Cleated |
| PPP-B-601 | Box, wood, cleated-Plywood |
| PPP-B-621 | Box, Wood Nailed and Lock-Corner |
| PPP-B-636 | Box, Shipping Fiberboard |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation-packaging, Methods of |
| MIL-S-901 | shockproof Equipment class HI (High Impact) Shipboard Application, Tests for |
| MIL-L-10547 | Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of |
| MIL-F-18870 | Fire Control Equipment, Naval Shipboard; General Specification for |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-H28 | Screw-Thread Standards for Federal services |

MILITARY

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-167-1 | Mechanical Vibrations of Shipboard Equipment (*Type I-Environmental and Type II-Internally Excited] |
| MIL-STD-648 | Design Criteria for Specialized Shipping Containers |



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MIL-STD-810 Evironmental Test Methods

DRAWINGS

NAVAL SEA Systems COMMAND (Code Ident 10001)

| | |
|---|---|
| LD 499687 | 60 Cycle 6W Synchro Servo Amplifier Assembly |
| SA 2530405 | Amplifier Test Fixture |

(Copies of specifications, standards, drawings, and publications requested by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer].

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Traffic Department, 1616 P Street, Washington, DC 20036).

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606].

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Preproduction sample. Unless otherwise specified in the contract (see 6.2), preproduction samples of the amplifier shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing amplifiers that comply with the technical requirements of the contract (see 4.3.1].

3.1.2 Materials. Unless otherwise specified in the contract (see 6.2], all materials and purchased parts entering into the final assembly of the amplifier shall conform to the detailed requirements given on the drawings and specifications listed in LD 499687 and on MI L-F- 18870.



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3.1.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for construction, and dimensions for the amplifier shall be in accordance with the applicable drawings, specifications, and publications listed on ID 499687.

3.1.4 Moisture, fungus, and corrosion. The amplifier shall be rendered moisture,fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications on LD 499687.

3.1.5 Marking. Markings of assemblies, subassemblies,and parts shall be in accordance with the applicable drawing LD 499687.

3.1.6 Threads. Unless otherwise specified, all threads shall be in accordance with FED-STD-H28. The class or fit for threads shall be as specified on the applicable drawings.

3.1.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.2 Definitions. The following definition is applicable to the environmental requirements of this specification:

3.2.1 Tropical enviroment. Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure which present an opportunity for fungi growth.

3.3 Performance requirements and product characteristics. The amplifier shall meet the following performance requirements and product characteristics when tested with the special test equipment specified in 4.4:

3.3.1 Voltage GAin. The voltage gain of the amplifier shall be such that when and indication on a vacuum tube voltmeter (VIVM) of the amplifier output is 22.0 ± O. 5 volts alternating current(Vac) the input signal to the amplifier is between 0.30 and 1.00 Vat.

3.3.2 Noise. The noise level of the amplifier with the input shorted shall be such as to produce a VIVM indication of less than 2.5 Vat.

3.3.3 Saturation. The output of the amplifier shall be between 32.0 and . Vac when the input signal indication is 60.0 * 0.5 Vac.

3.3.4 Power drain.

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3.3.4.1 Input Signal at 1.0 V, rotor shaft locked. The power drain of the amplifier shall be as not to produce a meter indicatiion greater than the fp;;pwomq wjem tje om[it sogma; tp tje a,[;ofoer 1.000 ± 0.025 Vac.

| Circuit identification (Vdc) | Power drain indication (mA, maximum) |
|---|---|
| 1 | 20.0 |
| 30 (A) | 28.0 |
| 30 (B) | 325.0 |
| 45 | 25.0 |

3.3.4.2 Input signal shorted, rotor shaft locked. The power drain of the amplifier shall be such as not to produce a meter indication greater than the following when the input signal to the amplifier is shorted:

| Circuit identification (Vdc) | Power drain indication (mA, maximun) |
|---|---|
| 1 | 6.0 |
| 30 (A) | 28.0 |
| 30 (B) | 10.0 to 80.0 |
| 45 | 20.0 |

3.3.5 Coarse-to-fine switching.

3.3.5.1 36-speed switch-coarse synchro, input signal 0.5 V level. The 36-speed coarse-to-fine switching action of the amplifier shall be such that when the input signal is 0.500 ± 0.025 V, the output signal shall not exceed 3.00 v.

3.3.5.2 36-speed switch-coarse synchro, input signal 2.0 V level. The 36-speed coarse-to-line switching action of the amplifier shall be such that when the input signal is 2.000 ± 0.025 V, the" outpt signal shall be 20.0 V minimum.

3.4 Environmental requirements. The amplifier shall meet the other applicable requirements of this specification during or after tests defined or specified in 4.6.

3.4.1 Vibration,. operating condition. The amplifier shall withstand the Type vibration requirement specified in MIL-STD-167-l; shall function during and after vibration tests; and, thereafter, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Vibration transportation. The amplifier, after being subjected to the tests specified in 4.6.2, shall show no damage and shall meet all the requirements of section 3.

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3.4.3 shock, nonoperating condition. 'She amplifier shall withstand the type C shock requirements specified in MIL-S-901 and, thereafter, shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.4 Shock: transportation, storage and handling. The amplifier after being subjected to the tests specified in 4.6.4, shall show no damage and shall meet all the requirements of Section 3.

3.4.5 Temperature , operating condition. The amplifier shall withstand operational ambient temperatures of between -18°C (0°F) and 85°C (185°F) in accordance with MIL-F- 18870; shall function during and after temperature tests; and, thereafter, shall show no damage or corrosion and shall meet all the performance requirements and product characteristics of this specification.

3.4.6 Temperature, nonoperating condition. The amplifier shall withstand storage temperatures of between -40°C (-40°F) and 71°C (160°F) and, thereafter, shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.7 Humidity. The amplifier shall withstand operational ambient humidities in accordance with MIL-F-18870 and, thereafter, shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.8 Tropical environment. The amplifier shall withstand the tropical environmental conditions specified in Method 508 of MIL-STD-810 and, thereafter, shall show no fungus growth, corrosion, or other damage, and shall meet all the performance requirements and product characteristics of this specification.

Prior to and after final assembly, all parts, components and the ass y shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be bid to neatness and thoroughness of soldering, marking of parts and assemblies, riveting, machine screw assemblage, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduct ion sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.



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## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapprovd by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements .

**4.2. Classification of inspections.** Inspection of the amplifiers shall be classified as follows:

a. Reproduction inspection (see 4.6)

b. Quality conformance inspection (see 4.5)

c. Periodic production inspection (see 4.6)

4.3 **sampling.**

of three amplifiers, subjected to the pre-production tests detailed in 4.6 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the amplifier by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk. Preproduction sample units shall be included in the quantity specified for delivery in the contract (see 6. 2).

**4.3.2 Periodic production sampling** Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspections of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sample units shall be included in the quantity specified for delivery in the contract (see 6. 2). Periodic production sampling shall be as follows :

1. Upon initiation Of production, randomly select one unit from each group of ten units. Upon satisfactory compliance of samples selected from three consecutive groups, shift to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples from three consecutive groups, shift to Step 3.

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Step 3. Randomly select one unit from each group of 50 UnitS. Continue with step 3 for the balance of the contract or until a defective unit is found.

When a defective unit is found in any group, the conatractor shall notify the procuring acativity Disposition of groups represented by failing samples shall be as specified by procuring activity (See 6.2 ). Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 Test equipment.The following test equipment is required to

4.4.1 Amplifier Test Fixture, Shop Accessory Dwg SA-2530405.

4.4.2 Power supplies

a. 208 Vat, 60 Hz, 100 watts (W), three phase, four wire

b. 200 Vat, 400 Hz, 100 W, three phase, four wire

C. 27 Vdc, 25 W

4. s Quality conformance inspection.

4. S. 1 Each amplifier delivered under contract shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection. Manufacturer's certification of materials specified in the applicable drawings used in lieu of material tests. Results of all tests shall be recorded on suitable data sheets and filed.

4.5.2 Test. conditions. Unless otherwwise specified, the amplifier shall be subjected to quality conformance tests under the following conditions :

4. 5.2.1 Temperature: Room ambient 18 to 35°C (65 to 95°F) .

4. 5.2.2 Altitude: Normal ground atmospheric pressure.

4.5.2.3 Humidity: Roan ambient to 95 percent relative maximum.

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4.5.3 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be *adequate in* quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.4 Test procedure. The amplifier shall be *tested in accordance with* the following detailed test procedure to detemine compliance with requirements of Section 3. The tests shall be perfomed in the order listed.

4. 5.4.1 Preparation for tests.

a. Plug Amplifier Test Fixture SA-2530405 into appropriate per outlets and set the switches and controls as follows:

| Switch or control | Position |
|---|---|
| AMPLIFIER SELECTOR | A6 |
| INPUT-OUTPUT CIRCUIT | 1 |
| TEST | Gain A |
| MOTOR CONTROL | F. F. Shorted |
| A3, A4, A5 CHANNEL | Channel 1 |
| INPUT SIGNAL | Med. |
| INPUT SIGNAL ADJUST | Fully Ccw |
| VTVM | Signal Out |
| RATIO TRANSFORMER | 1.00 |

b. plug the amplifier into receptacle chassis socket A6.

4. S. 4.2 Voltage gain test.

a. Turn MAIN POWER switch on.

b. Turn VIVM on (flat band].

Turn INPUT SIGNAL ADJUST clockwise (cw) until VTVM indicates an output of 22 v.

d. Turn VTVM switch to “Sig In” position. The reading shall be between 0.30 and 1.00 V.

4.5.4.3 Noise test.

a. Turn TEST switch to "Noise" position.

b. Turn VTVM switch to “Sig Out” position, The reading shall be less than 2.5 V.

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4.5.4.4 Saturation test.

a. Turn TEST switch to "GainA".

b. Set INPUT SIGNAL switch to "High" position.

c. nun VTVM switch to "Sig In" and adjust the input signal to 60 v.

d. Turn VTVM switch to "Sig out". The reading shall be between 32 and 43 V.

4.5.4. s Power drain indication.

a. Set INPUT SIGNAL switch to "Mod".

b. Turn VTVM switch to "Sig In" and adjust the input signal to one volt.

c. With the CIRCUT SELECTOR switch on "Main POwer Panel", measure and record the following currents, to meet the requirements of 3.3.4.1:

| Circuit identification (Vdc) | Power drain indication (mA, maximum |
|---|---|
| 1 | 20.0 |
| 30 (A) | 28.0 |
| 30 (B) | 325.0 |
| 45 | 25.0 |

d. Turn TEST switch to "Noise", and with CIRUIT SELECTOR switch on "Win Power Panel", measure and record the following currents, to meet the requirements of 3.3.4.2:

| Circuit identification (Vdc) | Power drain indication (mA, maximum) |
|---|---|
| 1 | 6.0 |
| 30 (A) | 28.0 |
| 30 (B) | 10.0 to 80.0 |
| 45 | 20.0 |

e. Turn CIRCUIT SELECTOR switch to "Off' position and set TEST Switch to "Gain A".

4.5.4.6 Coarse-to-fine switching.

a. Turn INPUT-OUTPUT switch to position 2.

b. Turn VTVM switch to "Sig In" and adjust the signal to 0.5 V.



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c. Change VTVM switch to "Sig (hat". The output signal shall not exceed 3.00 v to meet the requirements of 3.3. 5.1.

d. Turn VTVM Swtich to "Sig In" and adjust the signal to 2.00 V.

e. Change VTVM switch to "Sig Out". The output signal shall be not less than 20 V to meet the requirements of 3.3.5.2.

4.5.4.7 Visual and dimensional examination. Each amplifier shall be examined to verify conformance with the requirements for workmanship and dimensions.

4.5.4.8 Packaging inspection. The packaging, packing and marking shall be inspected to verify conformance with the requirements of section 5.

4.6 Preproduction and periodic production inspection. The Preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examination. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity as specified in the contract (see 6.2). The inspection data shall consist of the following:

a. Date of tests and names of persons conducting tests.

b. Run or test plan used.

c. Serial or identification number of unit.

d. Tests results of preproduction and periodic production tests.

4.6.1 Vibration, operating condition. The amplifier, rigidly mounted in a suitable vibration fixture, shall be subjected to vibration tests so as to verify coformance with the requirements specified in 3.4.1. The amplifier shall meet all the performance requirements and product characteristics of this specification during and after the vibration tests.

4.6.2 Vibration, transportation. Unless otherwise specified in the contract (See 6. 2) the amplifier, packaged and packed as specified in Section 5, shall be vibration tested as specified in MIL-STD-648.

4.6.3 Shock, nonoperating condition. The amplifier, rigidly mounted in a suitable shock future, shall be subjected to shock tests so as to verify conformance with the requirements specified in 3.4.3 and, thereafter, shall meet all the performance requirements and product characteristics of this specification.



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4.6.4 Shock; transportation, storage, and handling. Unless otherwise specified in the contract (see 6.2) the amplifier, packaged and packed as specified in section 5, shall be subjected to the appropriate shock isolation tests specified in MIL-STD-648.

4.6.5 Temperature, operating condition. The amplifier shall be subjected to temperature tests so as to verify conformance with the requirements specified in 3.4. S and, thereafter, shall meet all the performance requirements and product characteristics of this specification.

4.6.6 Temperature, nonoperating condition. The amplifier shall be subjected to temperature tests so as to verify conformance with the requirements specified in 3.4.6 and, therafter, shall meet all the performance requirements and product characteristics of this specification.

4.6.7 Humidity. The "amplifier shall be subjected to humidity tests so as to verify conformance with the requirements specified in 3:4.7 and, thereafter, shall meet all the performance requirements and product characteristics of this specification.

4.6.8 Tropical environment. The amplifier shall be exposed to fungus tests so as to verity conformance with the requirements specified in 3.4.8 and, thereafter, shall meet all the performance requirements and product characteristics of this specification.

4.7 Functional tests. At the conclusion of the tests specified in 4.6.2 and 4.6.4 functional and other tests, as applicable, shall be conducted on the amplifier to determine freedom from operational malfunction and compliance with all requirements of Section 3.

5. PACKAGING

S. 1 Preservation and packaging. Preservation and packaging shall be Level A *or c as* Specified in *the* contract (see 6. 2].

5.1.1 Level A. In addition to the following, Level A packaging shall be in accordance with MIL-E-17555.

5.1.1.2 Unit packaging. *unit packaging* of the amplifier shall be in accordance with Method IIe of MIL-P-116. The unit package shall contain one amplifier.

5.1.1.3 Intermnediate package. Not applicable.

5.1.2 Level B. Not applicable.

S. 1.3 Level C.



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5.1.3.1 cleaning, drying, and preservative application. Not applicable.

5.1.3.2 Unit packaging. Unit packaging of the amplifier shall be in accordance with Method III of MIL-P-116. The unit package shall contain one amplifier.

5.1.3.3 Intermediate package. Not applicable.

5.2 Packing. Packing shall be Level A, B or Commercial as specified in the contract (see 6.2].

5.2.1 Level A. In addition to the following, Level A packing shall be in accordance with MIL-E- 17555.

5.2.1.1 Exterior containers. Amplifiers shall be packed in a wood box in accordance with PPP-B-601, overseas type or PPP-B-621, Class- 2 (weight limit not to exceed 150 lbs.) or in a fiberboard box in accordance with PPP-B-636, Class 2 or PPP-B-591, overseas type (weight limit not to exceed 70 lbs. ).

5.2.2.2 Case liners. Wood exterior containers shall be provided with case liners in accordance with MIL-L-10547.

5.2.3 Commerical. Amplifiers shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity or immediate use Containers shall conform to Uniform Freight Classification, National Motor Freight Classification or to roles of other carriers applicable to the mode of transportation.

5.3 Levels A, B, and Commercial cushioning. The unit package or pack shall incorporate sufficient cushioning material, bracing, or other *adequate shock absorbing* devices to ensure *that the* amplifier will meet all the requirements of Section 3 after being tested in accodance with 4.6.2 and 4.6.4.
content of the cushioning material shall not exceed five percent..

5.4 Marking.

5.4.1 Special markings. None, unless otherwise specified (see 6.2).

**In addition to the markings required by the** contract, unit packages and shipping containers shall be marked in **accordance with the requirements of -STD-129, including serial numbers.**

6. NOTES



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6.1 Intended use. The 6-watt 60 Hertz synchro servo amplifier is intended for use in Angle Solvers Mk 18 Mod 2 and Mod 3, Fire Control System Mk 112 Mod 1, Sonar Receiving Sets AN/BQG-1, -2, -3, -4, or other suitable applications.

6.2 Ordering data. Procurment documents should specify the following:

a. Title, number, and date of this specification.

b. Any exceptions to this specification (see 3.1.2).

c. Required level of preservation and packaging (see 5.1).

d. Number of unit containers per pack.

e. Required level of packing (see 5.2).

f. Guarantee clause {see 6. S).

g. Assigned activity for preproduction and periodic production inspections and disposition of groups represented by failing samples (see 4. 3).

h. Quantity required (including preproduction and periodic production sample units (see 4.3) ).

i. That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed periodic production inspections.

j. When tests of 4.6.2 and 4.6.4 are waived (see 6.3).

k. Quantity required (including preproduction and periodic production sample units (see 4.3) ).

6.2.1 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Anneal Services Procurement Regulations, the data requirements identified helm will be developed as specified by an approved Data Item Description (DID) (DD) Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104. 9(n) " are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

| PARAGRAPH | DATA REQUIREMENT | APPLICABLE DID |
|---|---|---|
| 4.6 | Preproduction Inspection Report | DI-T-4902 |



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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer].

6.3 Waiver of packaged item tests. The tests specified in 4.6.2 and 4.6.4 may be waived by the procuring activity for Commercial packs only, provided the contractor guarantees safe delivery to the using activity (see 6.2.j).

6.4 The attention of the contracting officer is invited to the Quality Assurance provisions and options in MIL-P-116 and to 6.3 of this specification.

6.5 Guarantee. The contract shall include a requirement for a guarantee in accordance with 'the Armed Services Procurement Regulation (ASPR 1-324) stating that any amplifier which fails to meet the. requirements of this specification within one *year* after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - OS

Preparing Activity:
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Project No. 1290 -N343

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

**INSTRUCTIONS:** The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-A-22336B(OS), AMPLIFIER, SYNCHRO SERVO, 6 WATT 60 HERTZ

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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